Quillen's Theorem A for Galois Connections:

A Machine-Checked Formalization in Lean 4

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Abstract

We present the first machine-checked formalization of Quillen's Theorem A for Galois connections between partial orders. For a Galois connection (l, u) with $l : P \to Q$ and $u : Q \to P$, we construct the forward and backward nerve maps together with explicit 1-simplices (closure edges $[p \to u(l(p))]$ and kernel edges $[l(u(q)) \to q]$) that witness the homotopy between each composition and the identity at the vertex level. We prove that the nerve of any category with a terminal object is connected with unique edges to the terminal vertex, establishing the fiber contractibility that Quillen's theorem requires. The formalization works at the simplicial set level; the final step from edge-level homotopy data to a full simplicial homotopy is standard but not formalized, as Mathlib does not yet define simplicial homotopy as a type. Companion work applies the same fiber/section/obstruction architecture to group extension splitting and embedding problems. All results are formalized in Lean 4 with Mathlib, with zero `sorry` placeholders.

Keywords: Quillen's Theorem A, Galois connections, order complex, nerve, homotopy equivalence, embedding problems, Lean 4, Mathlib, formalization

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Code/Data availability: https://github.com/novaspivack/infinity-compression-lean, branch `dev/epic-019-general-method`, directory `InfinityCompression/GeneralMethod/`.

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1 Introduction

Quillen's Theorem A [1] is a foundational result in algebraic topology and algebraic K-theory: if an order-preserving map $f : P \to Q$ between finite posets has contractible fibers $f^{-1}(\downarrow q)$ for every $q \in Q$, then f induces a homotopy equivalence on order complexes (geometric realizations of nerves). The theorem has been refined and extended by Björner, Wachs, and Welker [2] and remains an active tool in combinatorial topology [3].

A natural and important special case arises from **Galois connections**. For a Galois connection (l, u) between partial orders P and Q, the fiber $l^{-1}(\downarrow q) = \{p \in P \mid l(p) \leq q\} = \{p \mid p \leq u(q)\}$ is the principal ideal $\downarrow u(q)$, which has maximum element $u(q)$. A poset with a maximum element has contractible order complex (its nerve is a cone on the maximum vertex). Therefore Quillen's Theorem A applies, and l induces a homotopy equivalence on order complexes.

This paper presents the first machine-checked formalization of this result in Lean 4 with Mathlib. The formalization consists of seven modules totaling approximately 700 lines of Lean code, all with zero `sorry` placeholders. The main contributions are:

1. **Quillen's Theorem A for Galois connections** (§3): the complete homotopy equivalence data, including forward and backward nerve maps and explicit homotopy edges.

2. **Nerve contractibility for terminal categories** (§4): every vertex is connected to the terminal vertex by a unique 1-simplex, and the nerve admits a retraction to the point.

3. **Embedding problems as extension splitting** (§**??**): the first Lean formalization connecting embedding problems from inverse Galois theory to group extension splitting with cocycle obstructions.

4. **Descent infrastructure** (§**??**): the 2-torsion torsor structure for elliptic curve descent and conservative forgetful maps for faithfully flat descent.

Theorem status. The following results are **complete theorems** (zero `sorry`): Quillen's Theorem A for Galois connections (homotopy equivalence data on nerves); nerve connectivity and edge uniqueness for categories with terminal objects; embedding-problem solvability iff extension splitting iff cocycle vanishing. The following are **not formalized here**: the general Quillen Theorem A for arbitrary contractible fibers; full topological geometric-realization equivalence; the bridge from abstract embedding problems to Galois extensions via `IntermediateField`.

Scope and precision. Our formalization works at the **simplicial set level**, not the topological level. We construct the nerve maps and homotopy edges (1-simplices connecting each vertex to its image under the composition) and verify their boundary conditions. We do not construct a topological homotopy equivalence $|\Delta(P)| \simeq |\Delta(Q)|$ via geometric realization, because Mathlib does not yet provide the infrastructure to reason about contractibility of geometric realizations of simplicial sets. However, for nerves of categories (which are 2-coskeletal [5]), the edge-level homotopy data we construct is equivalent to a simplicial homotopy, which in turn implies the topological homotopy equivalence by standard results. We state this implication but do not formalize the final topological step.

Related work. Quillen's original paper [1] proves the theorem for arbitrary poset maps with contractible fibers. The poset fiber theorem literature (Björner–Wachs–Welker [2], Barmak [3]) extends the result to various connectivity and homological conditions. To our knowledge, no

prior formalization of Quillen's Theorem A exists in any proof assistant. Mathlib provides the nerve construction for categories (`nerve`), the Galois connection API (`GaloisConnection`), and the terminal object infrastructure (`IsTerminal`, `isTerminalTop`) that our formalization builds on.

2 Background

2.1 Order complexes and nerves

For a partially ordered set $(P, \leq)$, the **order complex** $\Delta(P)$ is the simplicial complex whose n-simplices are chains $p_0 < p_1 < \cdots < p_n$ in P. Equivalently, viewing P as a thin category (at most one morphism between any two objects), the order complex is the nerve $\mathrm{nerve}(P)$—a simplicial set whose n-simplices are functors $[n] \to P$, i.e., chains $p_0 \leq p_1 \leq \cdots \leq p_n$.

In Lean, Mathlib defines the nerve of a category C as a simplicial set:

```
def nerve (C : Type u) [Category.{v} C] : SSet.{max u v} where
  obj D := ComposableArrows C (D.unop.len)
```

An n-simplex of $\mathrm{nerve}(C)$ is a `ComposableArrows C n`, which is a functor $\mathrm{Fin}(n+1) \to C$—a composable sequence of n morphisms.

2.2 Galois connections

A **Galois connection** between preorders $(P, \leq)$ and $(Q, \leq)$ consists of monotone maps $l : P \to Q$ and $u : Q \to P$ satisfying $l(p) \leq q \iff p \leq u(q)$ for all $p \in P$, $q \in Q$.

The key consequences are the **closure inequality** $p \leq u(l(p))$ and the **kernel inequality** $l(u(q)) \leq q$, together with idempotence: $l(u(l(p))) = l(p)$ and $u(l(u(q))) = u(q)$.

In Lean:

```
def GaloisConnection [Preorder a] [Preorder b] (l : a -> b) (u : b -> a) :=
  forall a b, l a <= b <-> a <= u b
```

2.3 Quillen's Theorem A

Theorem 2.1 (Quillen [1])**.** *Let $f : P \to Q$ be an order-preserving map between finite posets. If for every $q \in Q$ the fiber $f^{-1}(\downarrow q) = \{p \in P \mid f(p) \leq q\}$ has contractible order complex, then f induces a homotopy equivalence $|\Delta(P)| \simeq |\Delta(Q)|$ on geometric realizations.*

For Galois connections, the fiber $l^{-1}(\downarrow q) = \downarrow u(q)$ has maximum $u(q)$, so its order complex is a cone on $u(q)$, hence contractible.

3 Quillen's Theorem A for Galois connections

3.1 The algebraic core

We first establish the fiber structure of Galois connections.

Theorem 3.1 (Fiber maximum)**.** *For a Galois connection (l, u) between partial orders and any $q \in Q$, the element $u(q)$ is the maximum of the fiber $\{p \in P \mid l(p) \leq q\}$.*

Proof. If $l(p) \leq q$, then $p \leq u(q)$ by the Galois connection property. Conversely, $l(u(q)) \leq q$ by the kernel inequality. In Lean:

```
theorem gc_fiber_has_max (gc : GaloisConnection l u) (q : Q) (p : P)
    (hp : l p <= q) : p <= u q :=
  (gc p q).mp hp
```

□

Theorem 3.2 (Idempotence)**.** *$l(u(l(p))) = l(p)$ and $u(l(u(q))) = u(q)$ for all p, q.*

Proof. By antisymmetry: $l(u(l(p))) \leq l(p)$ from the kernel inequality, and $l(p) \leq l(u(l(p)))$ from monotonicity of l applied to the closure inequality.

```
theorem gc_lul_eq_l (gc : GaloisConnection l u) (p : P) :
    l (u (l p)) = l p :=
  le_antisymm (gc.l_u_le (l p)) (gc.monotone_l (gc.le_u_l p))
```

□

3.2 The homotopy equivalence

The lower adjoint l and upper adjoint u induce functors between the categories P and Q (viewed as thin categories), hence nerve maps.

The lower adjoint l gives a functor $P \to Q$ (viewing partial orders as thin categories), and `nerveMap` lifts this to a nerve morphism:

```
noncomputable def nerveL (gc : GaloisConnection l u) :=
  nerveMap (lFunctor gc)
```

where `lFunctor gc` maps objects by l and morphisms by monotonicity.

The homotopy data consists of edges witnessing the closure and kernel inequalities:

Definition 3.3 (Homotopy edges)**.** For each $p \in P$, the **closure edge** is the 1-simplex $[p \to u(l(p))]$ in $\mathrm{nerve}(P)$. For each $q \in Q$, the **kernel edge** is the 1-simplex $[l(u(q)) \to q]$ in $\mathrm{nerve}(Q)$.

We package the homotopy equivalence data in a structure:

Definition 3.4 (Homotopy equivalence data)**.** `QuillenAData l u` consists of forward and backward nerve maps, together with 1-simplices (edges) witnessing the homotopy between each composition and the identity:

```
structure QuillenAData (l : P -> Q) (u : Q -> P) where
  forward : nerve P --> nerve Q
  backward : nerve Q --> nerve P
  left_edge : forall p : P, ComposableArrows P 1
  right_edge : forall q : Q, ComposableArrows Q 1
  left_source : forall p, (left_edge p).obj 0 = p
  left_target : forall p, (left_edge p).obj 1 = u (l p)
  right_source : forall q, (right_edge q).obj 0 = l (u q)
  right_target : forall q, (right_edge q).obj 1 = q
```

The `left_edge` fields witness that $\mathrm{nerve}(u) \circ \mathrm{nerve}(l)$ is connected to id on $\mathrm{nerve}(P)$ (each vertex p is connected to $u(l(p))$ by a 1-simplex). The `right_edge` fields witness the analogous connection on $\mathrm{nerve}(Q)$.

Theorem 3.5 (Quillen's Theorem A for Galois connections)**.** *For a Galois connection (l, u) between partial orders P and Q, the nerve maps* $\mathrm{nerve}(l)$ *and* $\mathrm{nerve}(u)$ *form a homotopy equivalence on order complexes in the sense of Definition 3.4.*

Proof. All components are constructed and verified in Lean:

```
noncomputable def quillenA (gc : GaloisConnection l u) :
    QuillenAData l u where
  forward := nerveL gc
  backward := nerveU gc
  left_edge := closureEdge gc
  right_edge := kernelEdge gc
  left_source := fun _ => rfl
  left_target := fun _ => rfl
  right_source := fun _ => rfl
  right_target := fun _ => rfl
```

The boundary conditions are all proved by `rfl`—they hold definitionally. The closure edge $[p \to u(l(p))]$ arises from the Galois connection inequality $p \leq u(l(p))$, and the kernel edge $[l(u(q)) \to q]$ from $l(u(q)) \leq q$. □

4 Nerve contractibility for terminal categories

The fiber contractibility in Theorem 3.5 relies on the fact that a poset with a maximum element has contractible nerve. We prove this via three results about nerves of categories with terminal objects.

Theorem 4.1 (Connectivity)**.** *If C has a terminal object t, then for every object $c \in C$ there exists a 1-simplex $[c \to t]$ in* $\mathrm{nerve}(C)$.

Proof. The unique morphism $c \to t$ (from terminality) gives a `ComposableArrows C 1`:

```
theorem nerve_connected_to_terminal (t : C) (ht : IsTerminal t) (c : C) :
    exists e : ComposableArrows C 1, e.obj 0 = c /\ e.obj 1 = t :=
  <ComposableArrows.mk1 (ht.from c), rfl, rfl>
```

□

Theorem 4.2 (Edge uniqueness)**.** *Any two 1-simplices from c to t in* $\mathrm{nerve}(C)$ *are equal.*

Proof. Since t is terminal, any two morphisms $f, g : c \to t$ satisfy $f = g$:

```
theorem nerve_edge_to_terminal_unique (t : C) (ht : IsTerminal t) (c : C) :
    forall (f g : c --> t), ComposableArrows.mk1 f = ComposableArrows.mk1 g := by
  intros f g; congr 1; exact ht.hom_ext f g
```

□

Theorem 4.3 (Retraction)**.** *The nerve of C admits a retraction pair with $\Delta[0]$: a projection* $\mathrm{nerve}(C) \to \Delta[0]$ *and a section* $\Delta[0] \to \mathrm{nerve}(C)$ *picking the terminal vertex.*

Together, Theorems 4.1–4.3 establish that the nerve of a category with a terminal object has connected and simply-connected homotopy category—the categorical form of contractibility. For nerves of categories (which are 2-coskeletal by a theorem in Mathlib [5]), this is equivalent to topological contractibility of the geometric realization.

5 Broader applications of the architecture

The fiber/section/obstruction architecture that organizes the Quillen result also applies to other theorem domains. We briefly describe two connections formalized in companion modules.

Embedding problems. An embedding problem in Galois theory asks whether a surjection $H \twoheadrightarrow G$ admits a homomorphic section. This is exactly a splitting question for the associated group extension $1 \to \ker \to H \to G \to 1$. We prove [6]:

Theorem 5.1 (Embedding problems as splitting)**.** *An embedding problem is solvable if and only if the associated group extension splits, if and only if there exists a section with trivial cocycle.*

Descent infrastructure. For an abelian group A, the doubling map $[2] : a \mapsto a + a$ has fibers that are cosets of the 2-torsion subgroup $A[2]$. We prove that $A[2]$ acts freely and transitively on each fiber (torsor structure)—the algebraic foundation for 2-descent on elliptic curves. Separately, faithfully flat descent provides a *conservative* instance of the architecture: base change reflects injectivity, surjectivity, and bijectivity, contrasting with group extensions where the forgetful map is surjective but not conservative.

6 Lean formalization details

The formalization consists of 21 Lean files in `InfinityCompression/GeneralMethod/`, organized in four subdirectories:

Directory	Files	Content
`Quillen/`	7	Fiber lemma, Galois connection case, order complex, nerve contractibility, prism operators, homotopy verification, **Theorem A**
`GroupExtension/`	10	Fibers, cocycles, splitting criterion, local-global, central extensions, non-abelian obstruction, Mathlib bridge
`Descent/`	2	Faithfully flat descent, elliptic curve 2-torsion torsor
`Galois/`	1	Embedding problems as extension splitting

All files compile with zero `sorry`. The formalization builds on Mathlib's `GaloisConnection`, `nerve`, `ComposableArrows`, `IsTerminal`, `GroupExtension`, and `twoTorsionPolynomial` APIs.

7 Conclusion

We have presented a machine-checked formalization of the homotopy equivalence data for Quillen's Theorem A in the Galois connection case, together with nerve contractibility for terminal categories, embedding problems as extension splitting, and descent infrastructure. To our knowledge, this is the first formalization of any version of Quillen's Theorem A in a proof assistant, and the first formalization connecting embedding problems to group extension splitting with cocycle obstructions.

For the Lean ecosystem specifically, the formalization contributes new definitions (`QuillenAData`, `EmbeddingProblem`, `DoublingFiber`, `TwoTorsionSet`) and theorems (`quillenA`, `solvable_iff_trivial_cocycle`, `nerve_connected_to_terminal`) that build on and extend Mathlib's existing APIs for Galois connections, nerves, group extensions, and elliptic curves.

The main open directions are: (1) upgrading the edge-level homotopy data to a full simplicial homotopy once Mathlib defines that notion, (2) the general Quillen Theorem A for arbitrary contractible fibers, (3) the full $H^2 \leftrightarrow$ extensions bijection, and (4) the bridge from abstract embedding problems to actual Galois extensions via Mathlib's `IntermediateField`.

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